FORENSIC AUDIT EXECUTIVE REPORT

Dataset Evaluated: **clean_retail_data.csv** | Status: Complete

Phase 3 Baseline Score	Phase 4 Forensic Score	P3 Flagged	P4 Flagged	Newly Surfaced
31.75 / 100	13.00 / 100	38353	10900	5599

Forensic Test Breakdown

Test Executed	Calculated Risk Score	Records Flagged
Duplicate Detection	15.58	4706
Outlier Detection	10.41	6289

Executive Findings & Recommendations

Executive Audit Summary

1. **Executive Risk Assessment**
The system exhibits **High Risk** based on Phase 3 Baseline Risk Score (Score 1: 31.75/100) and Phase 4 Targeted Forensic Score (Score 2: 13/100). Both scores indicate significant vulnerabilities, with Score 2 reflecting critical gaps in forensic detection capabilities. The high number of flagged records (38,353 in Phase 3) and low overlap with Phase 4 (5,301) suggest systemic issues in risk prioritization and anomaly detection.

2. **Key Anomalies Identified**
- **Duplicate Detection**: 4,706 records flagged, indicating potential data redundancy or identity spoofing.
- **Outlier Detection**: 6,289 records flagged, highlighting irregular patterns in transaction behavior (e.g., inconsistent dates, missing values, or anomalous purchase volumes).
- **High-Risk Patterns**: Sample records show recurring issues such as "Bad" feedback, low ratings (1), and purchases in categories like clothing and groceries linked to suspicious activity. Missing date fields and mismatched geographic data (e.g., "Berlin" as a state) further compound risks.

3. **Actionable Recommendations for Audit Team**
- **Prioritize Duplicate and Outlier Analysis**: Focus on resolving duplicate records and investigating outlier patterns, particularly in transaction timestamps, geographic inconsistencies, and feedback anomalies.
- **Enhance Data Validation**: Implement stricter validation for critical fields (e.g., dates, phone numbers, addresses) to reduce false positives and improve data integrity.
- **Refine Scoring Models**: Re-evaluate Phase 4 forensic scoring to better align with Phase 3 baseline risks, ensuring targeted detection of high-impact anomalies.
- **Audit High-Risk Segments**: Intensify scrutiny of "Premium" and "Regular" customer segments, especially those with "Bad" feedback, low ratings, or unusual product categories.

This summary underscores the urgency of addressing systemic gaps in data governance and forensic detection to mitigate exposure to fraudulent or anomalous activity.